

You are given an array of m positive integers. These m values need to be inserted into a binary search tree (BST) in such a way that the sum of values of the leaves of the BST is minimized. Your task is to write a program to determine the sum of the leaf values when the above insertion is performed.

Read the input from STDIN and write the output to STDOUT. Do not write arbitrary strings while reading the input or while printing the output, as these contribute to the standard output.

Input Format:

The first line contains an integer, m , denoting the number of values in the array.
The second line contains m array elements separated by a single white space.

Output Format:

The output must be an integer denoting the sum of leaf values.

Constraints:

$$1 \leq m \leq 10^5$$

$$1 \leq A[i] \leq 10^5$$

Sample Input 1:

4
12 9 3 15

Sample Output 1:

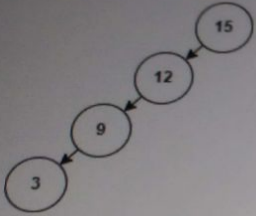
3

Explanation 1:

One possibility of adding the array elements in the tree is: first 15 is added, then 12, then 9 and at last 3. This results in the minimum sum of the leaf nodes, i.e. 3.

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One possibility of adding the array elements in the tree is: first 15 is added, then 12, then 9 and at last 3. This results in the minimum sum of the leaf nodes, i.e. 3.



Sample Input 2:

3

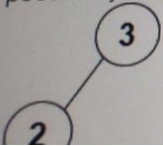
2 1 3

Sample Output 2:

1

Explanation 2:

One possibility of adding the array elements in the tree is: First 3 is added, then 2, then 1. This results in the minimum sum of the leaf nodes, i.e. 1.



Sample Input 2:

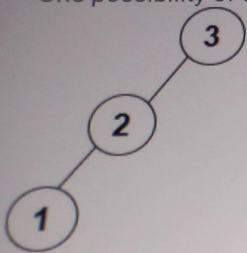
3
2 1 3

Sample Output 2:

1

Explanation 2:

One possibility of adding the array elements in the tree is: First 3 is added, then 2, then 1. This results in the minimum sum of the leaf nodes, i.e. 1.



Current Code

Previous Versions

Python 3

```
1 import heapq
2 # def sumOfLeafValues(array, m):
3 #     #res = 0
4 #     heapq.heapify(array)
5 #     total_sum=0
# array is array of m elements.
```

QUESTION: 56 GROUP: Coding SECTION: Coding 1

Mark(s): 10

Javed started Tour and travel company. He initiated several buses across various popular cities and advertised his business on all online travel platforms. Each day 2hrs before the Bus timing he gets the lists of all the online bookings from different websites, so somehow he managed to group all the data together and created a special code that consists of mobile number, gender, age, and seat number of the customers.

Now as per new online policies, he has to provide a supporting system to all senior citizens traveling on his bus. This supporting system covers special first aid kit and other necessary items.

To provide the proper supporting system to his passengers. Can you help Javed with a program by identifying the number of senior citizens traveling by his bus after processing the data provided?

Input data is in the following format

Input format:

[Mobile number][Sex][Age][Seat Number]

- Mobile number is 10 digits
- Sex occupies 1 character it can be M/F where M is male and F is Female
- Age occupies 2 digits
- Seat Number occupies 2 digits

Read the input from STDIN and print the output to STDOUT. Do not write arbitrary strings while reading the input or while printing, as these contribute to the standard output.

Constraint:

$1 \leq n \leq 44$

$| \text{strs} | = 15$

Input Format:

The first line of input determines the number of online bookings

The second line of input consists of n-number of lines with a string consisting of passenger details.

Output Format:

A single line of output consists of the number of senior citizens in the bus

Sample Input 1:

5

7868190130M2522

5303914400F3211

9272228200F4010

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QUESTION: 41 GROUP: Coding SECTION: Coding 2
Mark(s): 15

TIME LEFT: 00:39:28

Students of a college created flashcards f having all letters of the English alphabet as a part of an exercise on Teacher's day. The flashcards were created as follows.
 $f_2 = abc$, $f_3 = def$, $f_4 = ghi$, $f_5 = jkl$, $f_6 = mno$, $f_7 = pqrs$, $f_8 = tuv$, $f_9 = wxyz$

The students have to now form as many n -lettered words as possible from given n flashcards, by trying out all combinations of letters taking one from each of the mentioned flashcards, in the given sequence.

Write a program that accepts the number of flashcards to be used and what are flashcards to be used and print the possible words that can be formed with each word separated by space and end with letter 'X'.

Illustrate the above scenario by reading the Input from STDIN (standard input) and printing it to STDOUT (standard output) with a trailing newline.
You should not write arbitrary strings while reading the input and while printing as these contribute to the standard output.

Constraints:

- i. $n > 1$
- ii. The number of flashcards can have values as f_i where f_i ($2 \leq i \leq 9$).
- iii. The output should end with the letter X (in upper case)

Input Format:

The first line of input will be the number of letters in the word n .
The second line will be n flashcards, denoted by i from f_i , separated by a space.

Output Format:

In a single line with a space separation between each word, print all the possible words that can be formed by maintaining the order of characters in the flash card from left to right as the order provided in the input format. End the output with 'X'.

Sample Input 1:

2
5 6

Sample Output 1:

jm jn jo km kn ko lm ln lo X

Explanation 1:

f2 = abc, f3 = def, f4 = ghi, f5 = jkl, f6 = mno, f7=pqrs, f8 = tuv, f9 = wxyz

We need to form all possible 2 letter words, choosing 1 letter each from the 2 flashcards given 5 = jkl, 6 = mno.

The 1st combination takes the first letter from f5 as 'j', the second letter from f6 as 'm', is jm.

The 2nd combination takes the first letter from f5 as 'j', the second letter from f6 as 'n', is jn.

The 3rd combination takes the first letter from f5 as 'j', the second letter from f6 as 'o', is jo.

Similarly,

The 4th combination takes the first letter from f5 as 'k', the second letter from f6 as 'm', is km.

The 5th combination takes the first letter from f5 as 'k', the second letter from f6 as 'n', is kn.

The 6th combination takes the first letter from f5 as 'k', the second letter from f6 as 'o', is ko.

And,

The 7th combination takes the first letter from f5 as 'l', the second letter from f6 as 'm', is lm.

The 8th combination takes the first letter from f5 as 'l', the second letter from f6 as 'n', is ln.

The 9th combination takes the first letter from f5 as 'l', the second letter from f6 as 'o', is lo.

Hence, it will print " jm jn jo km kn ko lm ln lo X " as an output.

Sample Input 2:

3

2 3 4

Sample Output 2:

adg adh adi aeg aeh aei afg afh afi bdg bdh bdi beg beh bei bfg bfh bfi cdg cdh cdi ceg ceh cei cfg cfh cfi X

Explanation 2:

We need to form all possible 3 letter words, choosing 1 letter each from the 3 flashcards given 2=abc, 3=def, 4=ghi.

The 1st combination takes the first letter from f2 as 'a', the second letter from f3 as 'd', and the third letter from f4 as 'g', is adg.

The 2nd combination takes the first letter from f2 as 'a', the second letter from f3 as 'd', and the third letter from f4 as 'h', is adh.

The 3rd combination takes the first letter from f2 as 'a', the second letter from f3 as 'd', and the third letter from f4 as 'i', is adi.

Similarly,

by following the above pattern all possible combinations will be " adg adh adi aeg aeh aei afg afh afi bdg bdh bdi beg beh bei bfg bfh bfi cdg cdh cdi ceg ceh cei cfg cfh cfi X " .

Select Language

Previous Versions

Constraints:

- i. $n > 1$
- ii. The number of flashcards can have values as f_i where f_i ($2 \leq i \leq 9$).
- iii. The output should end with the letter X (in upper case)

Input Format:

The first line of input will be the number of letters in the word n .
The second line will be n flashcards, denoted by i from f_i , separated by a space.

Output Format:

In a single line with a space separation between each word, print all the possible words that can be formed by maintaining the order of characters in the flash card from left to right as per the order provided in the input format. End the output with 'X'.

Sample Input 1:

```
2
5 6
```

Sample Output 1:

```
jm jn jo km kn ko lm ln lo X
```

Explanation 1:

$f_2 = abc$, $f_3 = def$, $f_4 = ghi$, $f_5 = jkl$, $f_6 = mno$, $f_7 = pqrs$, $f_8 = tuv$, $f_9 = wxyz$

We need to form all possible 2 letter words, choosing 1 letter each from the 2 flashcards given $5 = jkl$, $6 = mno$.

The 1st combination takes the first letter from f_5 as 'j', the second letter from f_6 as 'm', is jm.

The 2nd combination takes the first letter from f_5 as 'j', the second letter from f_6 as 'n', is jn.

The 3rd combination takes the first letter from f_5 as 'j', the second letter from f_6 as 'o', is jo.

Similarly,

The 4th combination takes the first letter from f_5 as 'k', the second letter from f_6 as 'm', is km.

The 5th combination takes the first letter from f_5 as 'k', the second letter from f_6 as 'n', is kn.

The 6th combination takes the first letter from f_5 as 'k', the second letter from f_6 as 'o', is ko.

And,

The 7th combination takes the first letter from f_5 as 'l', the second letter from f_6 as 'm', is lm.

The 8th combination takes the first letter from f_5 as 'l', the second letter from f_6 as 'n', is ln.

The Abhaneri village stepwell has not been in use for a while. Stepwells are wells or ponds in which water is reached by descending a set of steps to the water level.

The stepwell in Abhaneri went out of use because the steps were very disorganised and people found it difficult to ascend/descend them. To correct this situation, the villagers decide to reconstruct the steps such that the heights of adjacent steps differ exactly by 1 unit.

The steps are in two directions, East-West and North-South. For symmetry, the villagers decide that the final steps in both directions must be identical. In each direction, the steps should be strictly descending, reach the water level, and then be strictly ascending. The lowest step, which is the water level, must have an equal number of steps to the left and to the right of it.

The existing N steps can be modified by reducing or increasing the height. Reducing or increasing the height of a step by 1 unit is counted as one change. Since funds are limited and stepwell construction requires many resources, the villagers want to make as few changes as possible.

Write a program to determine the minimum number of changes needed such that the villagers can rebuild their stepwell in the described way.

Read the input from STDIN and print the output to STDOUT. Do not write arbitrary strings while reading the input or while printing, as these contribute to the standard output and testcases will fail.

Constraints:

N is always an odd number

$3 \leq N \leq 10001$

$0 \leq \text{step height} \leq 10000000$

Input Format:

The first line of input contains N , the number of total steps in one direction (including the water level step).

The second line of input contains the steps in the East-West direction, separated by a single space.

The third line of input contains the steps in the North-South direction, separated by a single space.

Output Format:

The output contains the minimum number of total changes needed in both directions to reconstruct the stepwell.

Sample Input 1:

3

1 2 3

3 2 2

Sample Output 1:

2